CAR PRICE PREDICTION WITH MACHINE LEARNING

Technical Project Report

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Executive Summary

This project applies supervised machine learning regression algorithms to estimate used-car selling prices from vehicle characteristics. The workflow included data inspection and cleaning, feature engineering, preprocessing, train/test splitting, model comparison, hyperparameter tuning, and residual analysis. Four initial models were evaluated: Linear Regression, Decision Tree, Random Forest, and Gradient Boosting. Random Forest and Gradient Boosting were subsequently tuned using RandomizedSearchCV with five-fold cross-validation. On the reported held-out test split, Linear Regression achieved the lowest test RMSE (2.7120) and highest R^2 (0.7146), while tuned Gradient Boosting achieved the lowest test MAE (1.1845). Residual analysis revealed substantial errors, including a negative predicted price. Linear Regression is the current test-set leader by RMSE and R^2, but further validation and investigation of large errors are recommended before deployment.

1. Introduction

Used-car valuation is influenced by factors such as a vehicle's recorded present price, age, mileage, fuel type, transmission, selling channel, and ownership history. Machine learning can identify patterns in historical data and estimate selling prices for similar vehicles. This project builds and evaluates regression models for that purpose.

1.1 Problem Statement

Estimating a used car's selling price consistently is challenging because multiple vehicle attributes affect its value. A data-driven model may support estimates from available features, but reliability depends on data quality, representative examples, and appropriate evaluation.

1.2 Objectives

- Inspect and clean the used-car dataset.
- Explore relationships between selling price and vehicle attributes.
- Engineer features from vehicle age and mileage.
- Prepare numerical and categorical variables for regression.
- Train and compare four regression algorithms.
- Tune Random Forest and Gradient Boosting using cross-validation.
- Evaluate prediction errors and identify model limitations.

2. Dataset Description

The source file was named “car data.csv”. The initial dataset was reported to contain 301 rows and 9 columns. Two exact duplicate rows were identified during initial inspection; the final number of records after cleaning should be confirmed from the notebook output.

Feature	Description
Car_Name	Car name or model identifier
Year	Manufacturing year
Selling_Price	Target variable to predict
Present_Price	Recorded present price
Driven_kms	Distance driven in kilometres
Fuel_Type	Fuel category
Selling_type	Selling channel/category
Transmission	Transmission category
Owner	Previous ownership count/category

The price currency or unit was not confirmed in the available project information. Metrics are therefore reported on the dataset's original price scale without assigning an unverified currency.

3. Methodology

3.1 Data Cleaning and Inspection

The workflow inspected dimensions, data types, missing values, duplicates, invalid values, and potential outliers. Extreme observations were not assumed to be errors solely because they were unusual; they require verification before removal.

3.2 Feature Engineering

Car_Age was calculated as 2018 minus Year, using 2018 as the dataset reference year. Mileage_Per_Year was calculated as Driven_kms divided by Car_Age for positive ages. The reference year is historical and should not be interpreted as the current age of a vehicle.

3.3 Preprocessing and Data Split

Numerical features were imputed with the median and scaled using StandardScaler. Categorical features were imputed using the most frequent category and encoded with OneHotEncoder configured to ignore unseen categories. An 80/20 train/test split was created with random_state=42. Preprocessing was included in model pipelines during cross-validation to reduce leakage.

3.4 Algorithms

- Linear Regression: baseline model for estimating linear relationships between features and price.
- Decision Tree Regressor: models nonlinear relationships using tree-based decision rules.
- Random Forest Regressor: averages predictions from multiple decision trees.
- Gradient Boosting Regressor: builds sequential trees to reduce prediction errors.

3.5 Evaluation Metrics

Mean Absolute Error (MAE) measures average absolute prediction error. Root Mean Squared Error (RMSE) penalizes larger errors more heavily. R^2 measures the proportion of target variation explained relative to a mean-prediction baseline. Lower MAE and RMSE are preferred; higher R^2 is generally preferred.

4. Initial Model Results

The initial reported model comparison was:

Model	CV RMSE	Test MAE	Test RMSE	Test R^2
Linear Regression	1.5878	1.4342	2.7120	0.7146
Gradient Boosting	1.5740	1.2915	3.1271	0.6206
Decision Tree	2.2666	1.3608	3.2031	0.6019
Random Forest	1.5110	1.4633	3.5790	0.5030

Linear Regression achieved the lowest test RMSE and highest test R^2 in the initial comparison. Gradient Boosting achieved the lowest test MAE. Random Forest had the lowest initial CV RMSE but the highest test RMSE, showing that cross-validation and one test split can yield different rankings.

5. Hyperparameter Tuning

RandomizedSearchCV sampled 25 hyperparameter combinations for each of Random Forest and Gradient Boosting, using five-fold shuffled cross-validation on training data and negative root mean squared error as the scoring function. The best estimators were refitted on the full training set and evaluated on the held-out test set.

Model	CV RMSE	Test MAE	Test RMSE	Test R^2
Linear Regression baseline	—	1.4342	2.7120	0.7146
Gradient Boosting (Tuned)	1.3662	1.1845	2.9670	0.6584
Random Forest (Tuned)	1.5839	1.6303	4.0671	0.3582

Tuned Gradient Boosting reduced test RMSE from 3.1271 to 2.9670, approximately 5.1%, and improved test MAE from 1.2915 to 1.1845. Tuned Random Forest performed worse on this test split than its initial configuration. Linear Regression remained best on test RMSE and R^2. Its CV RMSE was not recalculated in the tuned comparison table, so its CV score cannot be compared directly there.

6. Residual Analysis

The Linear Regression residual plot showed many residuals clustered around zero, but also several substantial positive and negative residuals. A positive residual means the actual price exceeded the prediction (underprediction); a negative residual means the model overpredicted the price. The plot also included a negative predicted price, which is not realistic for a sale and illustrates a limitation of unconstrained Linear Regression.

These observations suggest that some vehicles are difficult to estimate. Possible explanations include influential observations, nonlinear relationships, omitted variables, and heterogeneous vehicle types. The largest-error records should be inspected before removing any observations.

7. Discussion

No model dominated every metric. Linear Regression led on test RMSE and R^2, while tuned Gradient Boosting led on test MAE and cross-validation RMSE among the tuned models. Model selection should consider validation stability and the cost of different prediction errors. Because the dataset is small and results depend on one train/test split, these test scores should be treated as development estimates rather than definitive evidence of future performance.

8. Limitations

- The dataset contains a relatively small number of observations.
- A single random split may produce variable performance estimates.
- Relevant factors such as vehicle condition, accident history, service history, and local market context may be absent.
- Linear Regression can produce unrealistic negative predictions.
- Extreme prediction errors can materially affect RMSE.
- The dataset currency/unit and current market representativeness were not verified.

9. Recommendations and Future Work

- Inspect and document the ten largest absolute prediction errors.
- Evaluate candidate models using repeated or nested cross-validation.
- Recalculate cross-validation scores for all models using the same folds and metric.
- Consider a log-transformed target if the price distribution is strongly right-skewed, evaluating predictions on the original scale.
- Compare additional algorithms and approaches that prevent negative predictions.
- Confirm dataset currency, units, and provenance.
- Evaluate on newer, independent listings before using predictions for real pricing decisions.
- Record dependency versions and save preprocessing together with the trained estimator.

10. Conclusion

The project demonstrated an end-to-end supervised learning workflow for used-car price prediction, including cleaning, feature engineering, preprocessing, model comparison, tuning, and residual analysis. Based on the current test results, Linear Regression is the leading candidate by RMSE and R^2, while tuned Gradient Boosting provides the lowest MAE and the best cross-validation RMSE among the tuned models. Large residuals and differences between validation and test results mean that further error analysis and robust validation are needed before declaring a final production model.

Appendix: Project Outputs

The notebook was configured to save outputs under car_price_outputs/, including cleaned and engineered datasets, model comparison CSVs, actual-versus-predicted and residual plots, prediction tables, and a serialized tuned pipeline. Confirm that each artifact exists before publishing the project.